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Measuring and managing pulmonary edema in ARDS: a Narrative Review

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Abstract

Despite significant advancements in the understanding of pulmonary vascular permeability and the preclinical development of compounds targeting pulmonary vascular permeability, their translation into clinical therapies for acute respiratory distress syndrome (ARDS) has remained unsuccessful. Among others, this translational gap can be attributed to limitations in measuring pulmonary vascular permeability in the clinical setting. This review aims to evaluate current and near-future modalities for quantifying pulmonary edema and their potential application in research settings.

We first outline the definition of ARDS and the pathophysiology of pulmonary edema, focusing on pulmonary vascular mechanisms and therapeutic targets to reducing vascular leakage. Next, we examine techniques for assessing pulmonary edema, including gravimetry (the preclinical gold standard), transpulmonary thermodilution (the clinical gold standard), as well as newer modalities such as lung ultrasound and chest CT. Finally, we discuss how pulmonary edema measurements may serve as meaningful endpoints in early-phase clinical trials.

The use of pulmonary edema as a primary endpoint in clinical trials may carry significant advantages, including more direct parameters for translation of pre-clinical studies on vascular leakage into clinical application, and possibly a higher success rate for transition to large phase III trials. Amidst the era of personalized medicine, the quantification of pulmonary edema holds promise in guiding clinical pharmacological trials for ARDS.

1. Introduction to ARDS

Acute respiratory distress syndrome (ARDS) is a syndrome characterized by protein-rich pulmonary edema throughout the lungs. The term ARDS was introduced by Ashbaugh et al. in 1967, after observing bilateral opacities on chest radiography accompanied by severe hypoxemia in 12 patients admitted to the ICU with diverse medical conditions(1). The definition of ARDS has changed over the years, with the latest update in January 2024(2). Despite changes in definition, three key criteria have remained consistent throughout the years. These include a) pulmonary edema that cannot be fully explained by a cardiogenic origin or fluid overload, b) acute onset of symptoms (< 7 days), and c) the presence of bilateral opacities on either chest radiography, computed tomography, or lung ultrasound. These common criteria stress the importance of acute, increased permeability, pulmonary edema in ARDS.

The latest guidelines include significant revisions: non-intubated patients are now considered within the ARDS definition, and arterial blood gas analysis is no longer required. These changes enable the diagnosis of ARDS in patients with early stage ARDS in the general ward, in patients not eligible for intensive care unit (ICU) admission, and in patients in a resource-limited setting where blood gas analysis is unavailable, which accounts for more than 80% of ARDS patients globally. Despite these advancements, limitations persist. The guideline definition still relies on parameters that are directly impacted by patient management, such as PaO₂/FiO₂ or SpO₂/FiO₂ ratios by positive end-expiratory pressure(3), and does not incorporate measurements of underlying pathophysiology(4). While these limitations are acknowledged, there are currently no feasible alternatives for either a clinical or a research definition(5).

One of the reasons to pursue uniform diagnostic criteria is the opportunity to standardize treatment for the patients. The pathophysiological mechanisms of hypoxemia, including pulmonary edema, pulmonary thrombi, and surfactant inactivation, are widely believed to remain consistent across all patients, regardless of the syndrome's cause (6). These mechanisms are potential therapeutic targets. However, to date, most ARDS treatments are supportive, with no direct therapies available that target these pathophysiological mechanisms(6). Supportive therapies include the use of lung-protective ventilation, neuromuscular blockade, and prone positioning, and have led to a significant reduction in mortality rates(7-10). Currently, research in ARDS focuses on personalized medicine to acknowledge and map heterogeneity among ARDS patients. An important example is the identification of hyper- and hypo-inflammatory subphenotype by Calfee et al.(11). Applying this subphenotyping in post-hoc studies has been shown to benefit patient-specific therapy targeting inflammation(12). Future prospective studies are needed to confirm these findings.

In comparison to inflammation and the immune response, targeting pulmonary permeability has gained relatively little attention in ARDS. In the past, few clinical studies have tested compounds directly targeting alveolocapillary integrity and vascular leak. Notably, the CounterCOVID trial provided evidence that such an approach can reduce mortality(13, 14). To our knowledge, only one study has targeted endothelial permeability and used pulmonary edema as a primary endpoint: the InventCOVID study using imatinib(15). Two other phase II trials, the BALTI and HARP trials, also used pulmonary edema as a primary endpoint(16, 17). The BALTI trial employed intravenous salbutamol to increase fluid clearance of pulmonary edema, while the HARP trial used statins to inhibit Hydroxyl-methylglutaryl coenzyme A reductase, based on the rationale that statins modify multiple processes, including the downregulation of TNF- α (18). Other trials have

tested compounds associated with endothelial permeability but used clinical or biological primary endpoints, such as adrenergic (19), and vitamin C(20). The lack of studies focusing on pulmonary vascular permeability might arise from the fact that pulmonary vascular permeability cannot be directly measured, and that surrogate or related outcome parameters are still difficult to interpret.

To assess the potential use of pulmonary vascular leak-reducing agents in future ARDS treatments, our review aims to explore current and emerging modalities for measuring pulmonary edema and the potential value of these approaches in future phase II trials. First, we discuss the pathophysiology of pulmonary edema in ARDS and its underlying biology. Next, we elaborate on the modalities available for measuring pulmonary edema, both currently and in the future. Finally, we examine how pulmonary vascular permeability and pulmonary edema can be utilized in clinical and research practice, both now and in the future.

2. Pathophysiology of pulmonary edema

2.1 Types and mechanisms of pulmonary edema

Pulmonary edema is the abnormal accumulation of fluid within the lungs and lung tissue outside the pulmonary vasculature. The increased interstitial and alveolar fluid is the net result of enhanced efflux of fluid across the capillary barrier versus impaired reabsorption and drainage(21). According to the Starling equation, which predicts fluid movement over a semi-permeable membrane, pulmonary edema is primarily categorized into two types: hydrostatic and hyperpermeability edema(22). Hydrostatic pulmonary edema occurs due to increased hydrostatic pressure predominantly in the capillary bed and the postcapillary venules, often associated with left-sided cardiac disease. The Starling equation dictates that such pressure drives fluid through the endothelial barrier into the interstitium(23). Throughout this process the endothelial barrier, especially the glycocalyx and adhesion junctions, stays intact, filtering out larger molecules such as albumins(24). (Figure 1A).

In contrast, hyperpermeability pulmonary edema is caused by the disruption of the endothelial and alveolar epithelial barrier. The resulting fluid is characterized by high protein content since the endothelial barrier no longer restricts the passage of larger molecules(25). Protein-rich pulmonary edema is a hallmark of ARDS and is often associated with widespread inflammation throughout the lungs, which causes diffuse alveolar damage(26). However, a large autopsy study found that only 45% of the patients who met the Berlin criteria for ARDS had diffuse alveolar damage (27, 28). In this study, the definition of DAD included the presence of a hyaline membrane observed under microscopy, indicating that not all patients who meet the ARDS criteria share the same histopathology.

While single pathophysiological mechanisms are often used to classify patients, these mechanisms often coexist and reinforce each other. This is frequently observed in ARDS, where both increased permeability and hydrostatic pulmonary edema can occur. The resulting increase in pulmonary microvascular pressure may be caused by pulmonary vasoconstriction and pulmonary thrombi(29).

In addition to excessive fluid inflow from increased hydrostatic pressure and permeability, fluid outflow may be compromised in ARDS as well. Two important structures in this context are the lymphatics and alveolar epithelium. The

lymphatic system is the main system to remove redundant fluid from the interstitium. Reabsorption from pulmonary edema back to the postcapillary venule, as suggested by the Starling equation, is minimal and less significant (Figure 1A)(23). As fluid flow increases, the lymphatic system compensates by enhancing drainage capacity. When drainage capacity is exceeded, the interstitium starts to overflow(30). Moreover, the reabsorption of fluid from the alveoli to the interstitium, primarily conducted by type I pneumocytes lining the alveolus, can be compromised. In ARDS, due to the severe inflammatory response, the epithelial cells necrotize, impairing their reabsorption capacity and exacerbating pulmonary edema(31). This review continues to focus on the endothelial barrier in the context of pulmonary edema, given the growing number of drugs targeting the endothelial barrier and the current limitations in quantifying lymphatic reabsorption capacity or alveolar destruction.

The endothelial layer is semi-permeable, allowing the transport of water and small molecules, while the alveolar layer is impermeable, preventing fluid accumulation within the alveolus(32). A canine study observed that the formation of pulmonary edema occurs in several stages. Initially, fluid enters the interstitium and is drained by the lymphatic system. Once the lymphatic drainage capacity is exceeded, interstitial edema develops. Once the interstitium reaches full capacity, fluid crosses the epithelial barrier resulting in alveolar flooding. In this context, it was observed that an alveolus is either completely filled with air or completely with fluid (33).

One of the biggest consequences of pulmonary edema is hypoxemia, explaining most of the respiratory failure seen in for example ARDS. Hypoxemia is caused by one of the following mechanisms: ventilation/perfusion (V/Q) mismatch, right-to-left shunt, diffusion limitation, low partial pressure of inspired oxygen, and alveolar hypoventilation(34). In ARDS, the main mechanisms of hypoxemia are shunt, based on a very low V/Q ratio (generally <0.01)(35). Similar observations were made in COVID-19 ARDS(36), in which computation models predicted hypoxemia to results from either hyperperfusion of nonaerated lung regions (alveolar flooding), or hypoperfusion of aerated regions (vascular injury and microthrombosis). Little support was found for the hypothesis that interstitial edema causes a diffusion limitation, which is supported by calculations that diffusion of inert gases is so quick, that no alveolar-endothelial gradient exists(37).

2.2 Biology of pulmonary vascular permeability

The endothelium is a single cell layer that lines all the blood vessels in the human body. Endothelial cells have a squamous shape oriented in the direction of blood flow(38). Recent molecular characterization studies have revealed that the alveolar endothelium exists of two subtypes of capillary endothelial cells. The majority are alveolar capillary endothelial cells which are involved in gas exchange, barrier function, and metabolism (called alveolar capillary cells, aCaps). A small minority of alveolar capillary endothelial cells are primarily involved in regeneration of the alveolar endothelium (called general capillary cells, gCaps) (39). The glycocalyx is a layer of proteoglycans and aminoglycosides(40), located at the luminal side of the endothelial monolayer. It plays a crucial role in communicating with the lumen and filtering macromolecules, maintaining a continuous oncotic pressure difference between the vascular lumen and the interstitium. The insights in the glycocalyx have significantly deepened our understanding in fluid movement as predicted by the Starling equation and our understanding of fluid reabsorption. Damage to the glycocalyx forms a central step

in the vascular permeability observed in ARDS (41). Endothelial cells signal to the environment among other via Weibel-Palade bodies, rod-shaped endothelial cell organelles, containing first-response inflammatory and homeostasis mediators. The main components of these granules are the von Willebrand factor and P-selectin(42, 43). Other proteins are interleukin 8, endothelin 1, eotaxin-3, osteoprotegerin, angiopoietin-2, and alpha-1,3-fucosyltransferase VI(44). When activated by inflammatory stimuli, endothelial cells can release the content of these granules on the cell surface(45). Interestingly, widespread damage to the alveolar endothelium and the underlying matrix is reflected by rises in circulating constituents of the glycocalyx, Weibel-Palade bodies and the extracellular matrix(46), indicating that these proteins could potentially be used as biomarkers of alveolar endothelial injury and ARDS.

Transport of fluid and molecules across the endothelial barrier occurs via either the paracellular or the transcellular route. Paracellular transport takes place between the cells through tight or adherens junctions. Under physiological conditions, the transport is passive and applies only to fluids and micromolecules. Transcellular transport is an active process that takes place via intracellular organelles, including caveolae and vacuoles, which move through the endothelial cell and allows the transport of macromolecules. This process is more often seen in fenestrated endothelial cells. These fenestrations are trans-endothelial channels that help in the filtration. Differences between endothelial cells depend on anatomical location, and whether the cell is arterial, venous, or capillary (47). Arterial and venous endothelial cells are larger, form an uninterrupted layer and possess a more substantial glycocalyx. Lung capillary endothelial cells are smaller, have a smaller glycocalyx, but are also uninterruptedly layered, and lack fenestrations. During inflammation, increased permeability allows the infiltration of larger proteins and cells, such as neutrophils. In the lungs, endothelial permeability is primarily initiated in the capillaries, whereas in other organs, it is initiated in the venules(48).

Adhesion proteins and the cytoskeleton are essential for maintaining endothelial barrier function. The cell-cell junctions include tight and adherens junctions. Tight junctions, located at the apical side, regulate water and molecule transport and maintain cell polarity(49). These junctions are multi-protein complexes; extracellularly, they bind to similar proteins on adjacent cells (homophilic), while intracellularly, they connect to actin filaments of the cytoskeleton. Key transmembrane proteins in tight junctions are claudins, occludins, and junctional adhesion molecules, with zonula occludens 1, 2, and 3 being important cytoplasmic proteins. Adherens junctions initiate and maintain cell-cell adhesion, with VE-cadherin as the primary transmembrane protein. Cadherins have cytoplasmic tails that bind to catenins, which connect to the cytoskeleton(50). (Figure 1B). In addition to changes in the permeability of the endothelial layer due to formation of intercellular gaps, advanced histology and tissue imaging studies have revealed widespread injury to the capillary network as a whole, characterized by severe endothelial injury and disrupted cell membranes, leading to loss of the capillary network (51, 52).

In the last decades, a number of preclinical ARDS studies have targeted pulmonary vascular hyperpermeability by enhancing the integrity of tight and adherens junctions, thereby improving endothelial barrier function. For instance, a study by Sanwal et al. demonstrated a beneficial effect of using localized claudin-5 to target tight junctions in an acute lung injury mouse model due to intratracheal E.coli(53). In 2021, Li et al. found BMP-9 as a potential endothelial protective factor, using Evans Blue in a mouse model(54). In this study, anti-BMP9 was associated with the loss of VE-cadherin junctions. Additionally, Aman et al. found that imatinib had a

protective effect on VE-cadherin junctions through the Arg Rac1 pathway(55). The tyrosine kinase inhibitor imatinib inhibits the Abl-related gene (Arg/Abl2) pathway, thereby stabilizing endothelial barriers and reducing permeability. In murine models, pretreatment with imatinib attenuated VEGF-induced vascular leakage in the skin and, in a sepsis model, significantly reduced lung and kidney edema(55).

Another promising target is Roundabout 4 (Robo4), an endothelial-specific protein from which the upregulation of ROBO4 is associated with suppression of TNF α induced endothelial hyperpermeability(56). Morita et al. found two SMAD pathways that regulated ROBO4, from which one positively and one negatively. They also found that an inhibitor of ALK1 enhances ROBO4 and consequently reduces pulmonary vascular permeability(57). (Figure 1B).

However, it remains to be determined, whether the dynamic opening and closing of the intercellular junctions sufficiently explains the massive vascular leak observed in the acute phase of ARDS. Ultrastructural analyses of lung tissue from deceased COVID-19 patients demonstrated extensive destruction of the capillary network, suggesting detachment of endothelial cells from the extracellular matrix(51) and turnover of the extracellular matrix component(46). These observations indicate that next to cell-cell junction integrity, cell-matrix interaction importantly determines endothelial barrier integrity(58, 59).

2.3 Summary and implications

In conclusion, understanding the formation and hypoxemic characteristics of pulmonary edema requires a clear grasp of how it develops and progresses, particularly how it infiltrates the interstitium and alveoli. While the distinction between hydrostatic and permeability pulmonary edema is well established, the roles of lymphatic drainage and alveolar epithelium are less known. Similarly, the mechanisms by which pulmonary edema induces hypoxemia—primarily through ventilation/perfusion (V/Q) mismatch and right-to-left shunting, rather than diffusion impairment—are not as widely recognized. Exploring how tight and adherens junctions contribute to the endothelial barrier offers valuable insights into potential therapeutic strategies for protecting or repairing the endothelial barrier, especially in cases of inflammation. These areas warrant further research to deepen our understanding and improve clinical research.

3. Current and future modalities to measure pulmonary edema

The principle, "If you cannot measure it, you cannot improve it", associated with Peter Drucker, holds significant relevance for pulmonary edema(60). Accurate measurements are critical to assessing its severity, evaluating its impact on clinical outcomes, and determining the reduction needed to improve patient health. To achieve these goals, parameters should ideally meet the following criteria: precise and reliable, widely available in both the ward and the ICU, and easy to obtain. In practice, each method for measuring pulmonary edema comes with its own strengths and limitations. This review explores current approaches to pulmonary edema assessment and examines emerging techniques that could shape its future evaluation.

3.1 Gravimetric

Gravimetry is considered the gold standard for measuring pulmonary edema; however, can only be performed post-mortem. As a result, its application is restricted to deceased patients or preclinical animal studies. Originally, gravimetry was a chemical method to determine the weight of a solid substance when

immersed in water. In the context of humans and animals, gravimetry has been adapted to quantify water content in specific tissues, first described in 1950 by Hemmingway(61, 62). For pulmonary edema, the lung is weighed directly after death, dried to remove all water, and weighed again. The difference between the wet and dry weights provides the amount of fluid, commonly referred to as the wet-to-dry ratio(63)(Figure 2). In gravimetry, it is important to correct for the amount of intravascular fluid. This amount is calculated by an intravascular tracer, most often hemoglobin. By determining the hemoglobin concentration, the fractional water content, and the density for both blood and homogenized lungs, the weight of lung blood can be calculated and subtracted from the total lung weight(64, 65).

3.2 Transpulmonary thermodilution

Transpulmonary thermodilution is the current clinical golden standard for measuring pulmonary edema in the ICU (Figure 2). Originally, this method utilized two indicators namely cold saline and a dye. However, the single-indicator method, using only cold saline, has since demonstrated comparable results while being more practical and cost-effective. Both methods rely on the principle that flow is derived from volume divided by time. Thus, the volume of a trajectory can be calculated by multiplying the flow by the mean time a substance takes to traverse this trajectory(66)(Figure 3A).

$$\text{Volume} = \text{mean Time} \times \text{flow}$$

To calculate pulmonary edema the volume of the lungs including the heart and pulmonary vasculature is subtracted from the volume of the heart and pulmonary vasculature. Given that the flow of venous return is approximately equal to cardiac output (approximately 5L/minute), the cardiac output can be utilized for flow calculations, with the trajectory extending from the right atrium to the left ventricle(67). A central venous line is inserted which approximates the right atrium and an arterial line to approximate the left ventricle (Figure 3A). In double indicator thermodilution, the dye is labeled to a large protein (albumin) or cell (red blood cell), preventing the dye from diffusing over the endothelial barrier(68). The cold saline, colder than 10°C, can pass through the endothelial barrier. The indicators are injected into the central venous catheter, passing the right atrium, right ventricle, lungs, left atrium, and left ventricle, to be detected in the aorta. The signal detected is either the concentration of dye in the blood or the difference in blood temperature. The mean time of the dye can be used to calculate the volume of the heart and pulmonary vasculature. The temperature of the cold saline also distributes to the lungs and can therefore be used to calculate the volume of the entire lung, heart, and vasculature (Figure 3 B1). The double indicator method is precise and well-validated as compared with gravimetry(69). However, its drawbacks include the use of a plasma-bound dye, making it cumbersome, expensive, and necessitating well-trained personnel.

Nowadays, single indicator thermodilution is most often used (PiCCO catheter)(70). In this method, two assumptions need to be made: 1) the exponential downslope time represents the volume of the lungs(70), and 2) the global end-diastolic volume multiplied by 1.25 represents the volume of the heart and pulmonary vasculature(71). The exponential downslope time denotes the period following a temperature drop during which the temperature most rapidly reverts to its baseline level. In a series of mixing chambers (heart, lung, heart) with a constant flow, the exponential downslope time represents the largest chamber, in this case, the lungs including its vasculature(71) (Figure 3A & B1). The global end-diastolic volume is linearly related to the volume of the pulmonary vasculature(70) and can therefore

thus be predicted (Figure 3 B2). Knowing the volumes of the other compartments, it is possible to calculate the fluid volume of the lungs, i.e., pulmonary edema (Figure 3 B3). The use of transpulmonary thermodilution is limited when the indicator trajectory is altered or when blood flow is inconsistent. Examples of an altered trajectory include massive pulmonary embolism or an intracardiac shunt. Inconsistent flow may occur in the presence of cardiac arrhythmias.

Two key parameters of transpulmonary thermodilution are the extravascular lung water index (EVLWi) and the pulmonary vascular permeability index (PVPI). EVLWi is directly derived from transpulmonary thermodilution and is defined as the amount of fluid inside the lungs but outside the vasculature and is indexed against the predicted body weight. PVPI is calculated by EVLW divided by pulmonary blood volume(72). An extensive review by Tagami et al found that EVLWi values of >10 mg/kg are representative of pulmonary edema and that PVPI values <2 are with certainty hydrostatic pulmonary edema whereas values >3 are with certainty hyperpermeability pulmonary edema(73).

In ARDS, EVLWi has moderate correlations with ventilator parameters like compliance and $\text{PaO}_2/\text{FiO}_2$, as well as lung injury scores(74). Significant differences in EVLWi have been observed between patients in different Berlin classifications and survivors and non-survivors(28, 75, 76). Currently, EVLWi is not often used in clinical decision-making. However, EVLWi can add value to fluid management because it has a significant correlation with fluid balance. Still, this has only been observed in a retrospective study(77). A large randomized trial found that the 60-day mortality between a conservative vs liberal fluid strategy is equal, however, there is an improvement in ventilator-free days and certain oxygenation parameters for the conservative fluid strategy(78). Furthermore, it is important to note that EVLWi cannot distinguish between hydrostatic and hyperpermeability pulmonary edema.

3.3 Lung ultrasound

Lung ultrasound is an easy-to-learn bedside technique with varying results for measuring pulmonary edema. Lung ultrasound relies on the interpretation of artifacts because ultrasound waves (1- 20 million hertz) cannot pass through the air(79). In a well-aerated lung, ultrasound waves are reflected at the pleura, creating horizontal A-lines. In cases of pulmonary edema, waves can pass through the fluid in the lung and are then reflected by air, creating vertical lines known as B-lines(80). Several scoring systems to quantify pulmonary edema have been developed, such as the lung ultrasound score, Global LUS score, and LUS-ARDS score (81). Counting B-lines is another common technique(81). A meta-analysis found that the lung ultrasound score (scoring at six areas per hemithorax) has a high specificity for diagnosing ARDS and moderate sensitivity. Furthermore, they found that the lung ultrasound score has a high specificity and sensitivity for distinguishing focal from non-focal ARDS(82). Lung ultrasound has potential roles in ventilator and fluid management of ARDS. Lung ultrasound can be used to evaluate the effect of alveolar recruitment in relation to positive end-expiratory pressure(83). For fluid management, lung ultrasound may be a safety measure against fluid overload(84).

For quantifying pulmonary edema lung ultrasound scores have been compared to transpulmonary thermodilution in various patient populations, with correlation coefficients ranging from approximately 0.4 to 0.9(85-95). Illustrative studies included those of Seibel et al. ($r= 0.59$) and Mayr et al. ($r=0.94$), both using unselected critically ill patients and extensive LUS protocols (26 and 28 sectors, respectively). The results of Mayr et al. suggested that lung ultrasound could be a

reliable non-invasive surrogate for transpulmonary thermodilution, while the results of Seibel et al. indicated that lung ultrasound might not be sufficient. There are distinct differences, Seibel et al. conducted a multi-center study with 184 patients with multiple LUS operators and interpreters, whereas Mayr et al. conducted a single-center study with 50 patients with one operator and interpreter. Previous studies have found an interobserver variability of $< 10\%$ for the interpretation of B-lines(96, 97), but little is known about inter-operator variability. Atmowihardjo et al. conducted another notable study comparing various LUS scores to transpulmonary thermodilution in a homogenous population of COVID-19 ARDS patients. This study found that all LUS scores showed similar correlation coefficients with transpulmonary thermodilution, but none exceeded 0.6 (95).

In conclusion, lung ultrasound forms an accessible means to detect pulmonary edema, as it is easy to learn, available at the bedside and inexpensive. Yet, due to the variable results in exact quantification pulmonary edema, lung ultrasound falls short as a precise surrogate of pulmonary edema at this moment. Furthermore, it cannot distinguish between hydrostatic and high-permeability pulmonary edema(Figure 2).

3.4 Chest computed tomography

While considerable research has focused on lung ultrasound, less is known about chest CT quantifying pulmonary edema. As with other modalities, there has been a transition to deep learning models with CT(98). Since 2001, seven studies, involving both animals and humans, have used several automated or semi-automated software systems to quantify pulmonary edema (99-105). The main principle for all systems is that the amount of ground glass opacity was quantified. The most recent studies used an artificial intelligence program trained on chest CTs from ARDS patients which derived the volumes of total opacity, ground-glass-opacity, and consolidation(106).

The chest CT quantification of pulmonary edema in these studies had correlation coefficients between 0.49 and 0.94 compared to gravimetry or transpulmonary thermodilution. The lower correlation coefficient ($r=0.49$) was found in an undifferentiated ICU population(102), while the higher coefficients were found in an ARDS population ($r=0.91, 0.72, 0.71$)(100, 103, 105). These studies present promising results, especially for ARDS patients, but the lack of large prospective studies makes it difficult to draw robust conclusions.

A pronounced advantage of chest CT is that it can be standardized, is available in both the ward and ICU settings, and is non-invasive. However, discrepancies exist between the studies, due to differences in CT scanning protocols. Variations include the use of breath holds, scanning at end-inspiratory or end-expiratory phases, and whether patients were mechanically ventilated or breathing spontaneously. Another factor for potential overestimation is that most deep-learning models have not yet been corrected for vascular volume(106) (Figure 2).

Looking ahead, chest CT may become a viable modality for measuring pulmonary edema in patients before intubation. Yet, several questions need to be addressed, including the chest CT protocol that is optimal for measuring pulmonary edema, how different patient populations affect the measurement, and the use of deep learning models to correct for pulmonary vasculature volume.

3.5 Chest radiography

Chest radiography is an easily obtainable and inexpensive imaging modality that is routinely used in both the ward and the ICU. The RALE score is commonly applied to assess the extent of pulmonary edema on chest radiographs. The scoring system divides the chest X-ray into four quadrants and assigns a score to each quadrant based on the extent of consolidation and the density of opacification. Overall, the RALE score demonstrates good inter-rater reliability(107, 108).

One study correlated the RALE score with gravimetry, the gold standard for measuring pulmonary edema(107). This analysis included 72 potential donor lungs that had been declined for transplantation and demonstrated a significant correlation, with a coefficient of 0.59. Another study compared the RALE score with transpulmonary thermodilution in 65 patients with COVID-19-related ARDS(105). In contrast, this study found no significant correlation ($r = 0.15$, $P = 0.3$). However, the analysis was retrospective and relied on a single interpreter for RALE score assessment, which may have limited the reliability of the findings(105).

Several studies have evaluated the association between the RALE score and clinical outcomes, with contrasting outcomes (107, 109-114). High RALE scores were independently associated with lower $\text{PaO}_2/\text{FiO}_2$ ratios and increased 90-day mortality in a cohort of 174 patients with ARDS(107). In line with this, baseline RALE score and the change in RALE score after the onset of ARDS were significantly associated with 90-day mortality(113). In a cohort of 425 patients with COVID-19 ARDS, a strong association was found between RALE scores and respiratory mechanics, gas exchange parameters, and plasma biomarkers(109). In contrast, Kotok et al. did not observe an association between RALE scores and survival in a cohort of 129 patients (including 108 with ARDS), although they did find an association with the duration of mechanical ventilation(108). Similarly, no association was found between RALE scores and either survival or ventilator-free days in a cohort of 139 patients with COVID-19-related ARDS(114).

Overall, limited evidence is available regarding the use of the RALE score as a quantitative measure of pulmonary edema, and the initial studies have yielded inconclusive results. While the RALE score has shown potential as a predictor of survival and clinical progression, findings across studies remain inconsistent.

3.6 Blood biomarkers

The value of blood biomarkers has become increasingly important in current medicine, and also in ARDS research and care blood biomarkers have an important place to personalize medicine. Examples include the use of biomarkers to classify patients as hyper- and hypo-inflammatory(11). Biomarkers may also be used to distinguish direct and indirect ARDS, in which direct ARDS showed higher expression of epithelial biomarkers, whereas indirect ARDS is characterized by more pronounced expression of endothelial markers(115). Due to the heterogeneity of ARDS, there is no single biological biomarker that can predict or evaluate therapeutic response(116). A meta-analysis found that the endothelial markers angiopoietin-2 and soluble receptor for advanced glycation-end products (sRAGE) were associated with the development of ARDS but did not find markers that associated with the outcome(117). This is in line with a study that found that RAGE was one of the most significant proteins to predict critical illness in COVID-19. This study also revealed major disturbances in pathways of cell adhesion and extracellular matrix organization in critically ill patients during the first three days

of admission(46). Vascular endothelial (VE)-cadherin is an important molecule adherens junctions and thereby contributing to endothelial barrier integrity. Studies found that in septic patients fragments of VE-cadherin i.e. soluble VE-cadherin represent endothelial barrier damage and thus may be used as a diagnostical tool of barrier damage(118).

One study has examined the correlation between biological biomarkers, pulmonary edema and, pulmonary vascular permeability in ARDS using transpulmonary thermodilution. This study found that no single biomarker can accurately estimate these parameters. However, pathway analysis has shown a close relationship between acute-phase inflammatory pathways and pulmonary vascular permeability(105) (Figure 2). Large scale proteomic platforms may guide future identification of specific pathways and individual biomarkers for diagnosis and predicting prognosis and treatment response, although large patient numbers are required.

3.7 Other modalities

Some modalities have shown good or promising results in quantifying pulmonary edema; however, they are currently not used because of impracticalities and the availability of better alternatives. As mentioned earlier, double-indicator thermodilution is rarely used because single-indicator thermodilution is easier and provides similar results(119, 120).

MRI, which uses magnetic fields and radio waves to align protons to create images, is particularly useful for soft tissues and can also quantify pulmonary edema. Studies from the 1980s and the 1990s showed good results compared to gravimetry in preclinical animal models(121, 122). However, no validation has been performed in ARDS patients, likely because of the impracticalities of the magnetic field with MRI in ventilated critically ill patients, especially before 2000(123, 124).

Lung scintigraphy is a method that most commonly uses radiolabeled albumin to determine the leakage of the pulmonary capillary barrier. In the case of pulmonary vascular permeability, a portion of the tracer is absorbed into the lung interstitium, while the remainder remains in the circulation. The part of the tracer in the interstitium is more concentrated and therefore provides a stronger signal(64). Currently, scintigraphy is not commonly used to assess pulmonary edema due to the availability of more precise measurements that do not necessitate radioactive tracers.

PET-CT studies have shown varying results, mainly in animal models and heart failure patients(125, 126). This technique involves the use of PET-CT combined with two tracers, which complicates its application in clinical settings. The most commonly used tracers are a continuous infusion of ^{15}O -labeled water ($^{15}\text{O}\text{-H}_2\text{O}$) and inhaled ^{11}C -labeled carbon monoxide (^{11}CO). $^{15}\text{O}\text{-H}_2\text{O}$ is highly permeable and can serve as a proxy for total intrathoracic water volume, while ^{11}CO binds to hemoglobin and is used to estimate total intravascular blood volume. The EVLW is then calculated by subtracting the intravascular volume from the total water volume.(125, 126). Although these modalities have potential, practical limitations and the emergence of better alternatives have limited their application in research and clinical practice.

3.8 Conclusions

Currently, there is no ideal modality for estimating pulmonary edema (Figure 2). Although transpulmonary thermodilution measurements are precise, they are invasive and only applicable in the ICU. Gravimetry can only be used post-mortem and can therefore only be used as a reference measure. This may be one of the reasons why measuring pulmonary edema is underexplored; therefore, it is rarely used in current clinical and research practice. Nevertheless, the future perspectives are promising. In particular, improvements in imaging techniques, such as chest CT, hold promise for a non-invasive measure applicable on the ward. A combination of techniques may provide sufficient diagnostic power and could potentially result in high sensitivity and specificity.

4. Designing the optimal phase II trial to target lung vascular leak

Emerging evidence supports the value of using pulmonary edema as an endpoint in future phase II clinical trials, as summarized in this review. Despite being a hallmark of ARDS, pulmonary edema remains underutilized in clinical practice and underrepresented in contemporary clinical research(1). Numerous preclinical studies on acute lung injury have employed pulmonary edema as an endpoint, demonstrating its reliability in assessing therapeutic efficacy and highlighting pulmonary vascular permeability as a viable therapeutic target(127, 128). Among the three clinical trials that have used pulmonary edema as a primary endpoint, results have varied, suggesting its potential to differentiate treatment responses(15-17). Importantly, pulmonary edema has been consistently associated with mortality, clinical status, and ventilator outcomes, underscoring its relevance as a meaningful and clinically informative parameter(73).

Designing an optimal phase II clinical trial requires careful consideration of several factors: the timing of intervention, the patient population, outcome parameters, and the choice of therapeutic targets (Figure 4). With respect to timing, it is essential to determine whether an intervention is intended to prevent edema formation or to accelerate its resolution. Data from studies utilizing transpulmonary thermodilution suggest that pulmonary edema typically peaks by the time of intubation and does not substantially increase thereafter during ICU admission(15, 16). These findings imply that most of the edema accumulation occurs before intubation. Therefore, therapies aimed at preventing edema, possibly those enhancing endothelial barrier function, may be most effective when administered early, at the onset of edema formation. Studies testing these therapies would need to be conducted on the ward in pre-ICU patients. As transpulmonary thermodilution is not available at the ward, chest CT may emerge as a suitable alternative for assessing pulmonary edema in this context. Treatments designed to promote edema resolution may be more appropriate after intubation, when the injury is already established. In light of the findings of Ackermann et al, also regeneration of the lost capillary bed might form a relevant therapeutic target(51).

In addition to timing, patient heterogeneity plays a key role in treatment response. Personalized approaches are becoming increasingly important in clinical trial design. In ARDS, distinct hyper-inflammatory and hypo-inflammatory subphenotypes have been identified. Evidence suggests that these groups respond differently to targeted therapies(12). Interestingly, also subphenotypes may be time- dependent, as recent studies demonstrated a shift from hyper- to hypo-

inflammatory subphenotype over time (129). However, the relationship between the subphenotypes and pulmonary edema burden remains poorly explored. Notably, De Brabander et al. identified a subgroup of patients with elevated IL-6, TNFR1, and SP-D who responded favorably to imatinib, a drug that reinforces the pulmonary endothelial barrier(130). In a subsequent trial, this same subgroup demonstrated a specific reduction in pulmonary edema following treatment with imatinib. These findings suggest that patients exhibiting higher levels of endothelial or epithelial injury may benefit more from therapies that modulate vascular permeability(15). Additionally, pulmonary causes of ARDS are associated with greater pulmonary edema and PVPI burden compared to extrapulmonary causes. This raises the possibility that certain subgroups may be more responsive to therapies specifically targeting pulmonary edema(131).

An ideal outcome parameter for assessing pulmonary edema should be precise, valid, pathophysiologically relevant, non-invasive, and feasible for longitudinal monitoring(132). Ideally, such a parameter would approximate the gold standard across the full spectrum of edema severity, accurately reflecting damage to the alveolar-endothelial barrier. To better track the evolution of pulmonary edema and identify optimal therapeutic windows, longitudinal sampling is essential. However, invasive techniques for quantifying edema are largely confined to ICU settings, leaving significant gaps in understanding its onset and progression in non-ICU patients. A non-invasive parameter would help overcome these limitations. As this review highlights, such a parameter is not yet available. Nonetheless, emerging tools, including chest CT imaging and plasma biomarkers, offer promise in bridging this gap. For future phase II trials, it will be crucial to prospectively evaluate these novel approaches in comparison with established techniques such as transpulmonary thermodilution.

Lastly, selecting the appropriate therapeutic target is critical. In the context of preserving or restoring the pulmonary endothelial barrier, key targets include direct mechanisms such as strengthening cell-cell and cell-matrix junctions, as well as more indirect mechanisms, including immune modulation and promotion of endothelial regeneration(133, 134). These approaches involve distinct pathophysiological pathways, operate over different temporal time windows, and have divergent impacts on pulmonary edema—either preventing its formation or accelerating its resolution.

In summary, pulmonary edema is a central feature of ARDS and a promising outcome parameter for phase II clinical trials. Its pathophysiological relevance, clinical associations, and potential for measurement, particularly through novel non-invasive techniques, position it as a valuable target for advancing therapeutic development. Optimizing its use will require attention to timing, patient stratification, outcome measurement, and therapeutic target (Figure 4).

Future perspectives

In conclusion, the quantification of pulmonary edema holds promise for bridging pre-clinical findings with large-scale clinical trials. Emerging non-invasive modalities provide novel insights, particularly in detecting the stages before ARDS, potentially opening a new therapeutic window for preventing pulmonary edema formation. This may also be facilitated by earlier recognition of ARDS following the

642 new global guidelines. Looking ahead, the availability of pulmonary edema-
643 reducing agents may further enhance these efforts. Quantification of pulmonary
644 edema not only facilitates the initiation of targeted therapies but also enables the
645 evaluation of treatment responses in the future.

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List of abbreviations

ARDS	Acute respiratory distress syndrome
EVLWi	Extravascular lung water
ICU	Intensive care unit
PVPi	Pulmonary vascular permeability index
ROBO 4	Roundabout Guidance Receptor 4
sRAGE	Soluble receptor for advanced glycation-end products
TNF- α	Tumor necrosis factor alpha
V/Q	Ventilation/perfusion
VE	Vascular endothelial

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JS conceived the project, performed the literature review, drafted images, wrote the manuscript. JA conceived and supervised the project. All authors contributed to the writing, critically revised the manuscript and gave approval of the final version to be published

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